



System Documentation

Electronic Equipment
ECU 7 (ADEC)
with SAM and EMU 7
12V 4000 Px3
16V 4000 Px3
20V 4000 Px3
Application: Oil & Gas Offshore

E532458/02E

Rolls-Royce Solutions refers to Rolls-Royce Solutions GmbH or an affiliated company pursuant to § 15 AktG (German Stock Corporation Act) or a controlled company (joint venture), and Rolls-Royce Solutions Ruhstorf GmbH.

© Copyright Rolls-Royce Solutions

This publication is protected by copyright and may not be used in any way, whether in whole or in part, without the prior written consent of Rolls-Royce Solutions. This particularly applies to its reproduction, distribution, editing, translation, microfilming and storage and/or processing in electronic systems including databases and online services.

All information in this publication was the latest information available at the time of going to print. Rolls-Royce Solutions reserves the right to change, delete or supplement the information provided as and when required.

Table of Contents

1	Safety			
1.1	Important provisions for electronic products	6	3.1.3.14.2	Transistor output TOP 2
1.2	Intended use of electronic products	7	3.1.3.14.3	Transistor output TOP 3
1.3	Personnel and organizational requirements	8	3.1.3.14.4	Transistor output TOP 4
1.4	Safety regulations for commissioning and operation of electronic products	9	3.1.3.15	Analog output for display instruments
1.5	Safety regulations for assembly, maintenance, and repair work on electronic products	10	3.1.3.16	Digital inputs
1.6	Fire and environmental protection, consumables and auxiliary materials for electronic products	12	3.1.4	Redundant Engine Control Unit
1.7	Standards for warning messages in the text and highlighted information	13	3.1.4.1	Technical Description
2	System Layout		3.1.4.1.1	Redundant engine management system with ADEC (ECU 7)
2.1	General information and fully integrated system design (excluding optional auxiliary and starting systems)	14	3.1.4.1.2	Installation location of the redundant engine management system
2.2	System design of an Engine Control System ECS featuring Engine Control Unit ECU 7	18	3.1.4.1.3	System - Overview
2.3	System design of an Engine Control System ECS with redundant Engine Control Unit ECU 7	20	3.1.4.1.4	Switch Box (SBX)
2.4	Bus topology	22	3.1.4.1.5	System description and system boundaries
2.5	Scope of supply	23	3.1.4.1.6	Functional requirements
3	Devices and Functions		3.1.4.1.7	System, system context and system boundaries
3.1	Engine Governor ECU 7 (ADEC)	24	3.1.4.1.8	Switching criteria
3.1.1	Engine Control Unit ECU 7 - Function	24	3.1.5	Field Data Handling
3.1.2	ECU 7 - Technical data	26	3.2	Engine Monitoring Unit EMU
3.1.3	Control and Monitoring Functions of Engine Control Unit ECU 7	28	3.2.1	Engine Monitoring Unit EMU 7 - Function
3.1.3.1	Control functions	28	3.2.2	EMU 7 - Technical data
3.1.3.2	Engine start	29	3.2.3	Monitoring Functions of Engine Monitoring Unit EMU
3.1.3.3	Engine stop	31	3.2.3.1	Measuring point limit monitoring
3.1.3.4	Emergency stop (ESI)	32	3.2.3.2	Engine speed monitoring
3.1.3.5	Closed-loop control functions	33	3.2.3.3	Individual exhaust temperatures - Monitoring
3.1.3.6	Setpoint speed generation	34	3.3	Service and Automation Module SAM
3.1.3.7	Speed control	37	3.3.1	Service and Automation Module SAM
3.1.3.8	Fuel quantity during starting	38	3.3.2	SAM - Technical data
3.1.3.9	Idling speed governor - maximum-speed governor - fuel injection governor	41	3.3.3	Control and Monitoring Functions of Service Automation Module SAM
3.1.3.10	Common Rail injection system	42	3.3.3.1	Cylinder cutout disabled
3.1.3.11	Safety features	43	3.3.3.2	LO coolant level HT and NT (optional from SAM)
3.1.3.12	Engine protection functions	44	3.3.3.3	Start interlock
3.1.3.13	ECU stop status	46	3.3.3.4	Overspeed alarm
3.1.3.14	Transistor Outputs	47	3.3.3.5	SAM - Alarm handling
3.1.3.14.1	Transistor output TOP 1	47	3.3.3.6	Water in fuel prefilter (optional)
			3.3.3.7	HI coolant temperature
			3.3.3.8	LO Lube oil pressure
			3.3.3.9	HI Crankcase pressure
			3.3.3.10	Fuel leakage
			3.3.3.11	HI Lube oil temperature
			3.3.3.12	HI Exhaust temperature
			3.3.3.13	LO ECU voltage
			3.3.3.14	Differential oil pressure
			3.3.3.15	Differential pressure fuel prefilter (optional)
			3.3.3.16	Generator temperature monitoring
			3.3.3.17	SAM binary outputs - Additional settings
			3.3.4	Engine speed
			3.3.5	Exhaust temperature, A side

3.3.6	Exhaust temperature, B side	118
3.3.7	Lube oil temperature	119
3.3.8	Crankcase pressure	120
3.3.9	Intake air temperature	121
3.3.10	Fuel temperature	122
3.3.11	Coolant temperature	123
3.3.12	Lube oil pressure	124
3.3.13	Engine coolant pressure after pump	125
3.3.14	Intercooler coolant pressure	126
3.3.15	Fuel pressure	127
3.3.16	Charge-air pressure	128
3.3.17	Fuel pressure (common rail)	129
3.3.18	Turbocharger speed	130
3.3.19	Charge-air temperature	131
3.3.20	Intercooler coolant temperature	132
3.3.21	Injection quantity	133
3.3.22	T- Winding U	134
3.3.23	T- Winding V	135
3.3.24	T- Winding W	136
3.3.25	T-Bearing DE	137
3.3.26	T-Bearing NDE	138
3.3.27	T-Auxiliary 1	139
3.3.28	T-Auxiliary 2	140
3.4	Emergency Air-Shutoff Flap Control	141
3.4.1	Emergency air-shutoff flap control - Introduction and overview	141
3.4.2	Function	142
3.4.2.1	Normal operating state (emergency air- shutoff flap open)	142
3.4.2.2	Shutoff state (both emergency air-shutoff flaps closed)	143
3.4.2.3	Reset after closing the emergency air-shutoff flap	144
3.4.2.4	Emergency air-shutoff flap control (Rolls- Royce Solutions option only)	145
3.4.2.5	Emergency air-shutoff flap - Monitoring	148
3.4.2.6	Assignment parameters	157
3.4.3	Mechanical Assemblies of the Emergency Air- Shutoff Flaps	158
3.4.3.1	Emergency air-shutoff flap - Location	158
3.4.3.2	Emergency air-shutoff flaps (ASO)	160
3.4.3.3	Emergency air-shutoff flap - Coil, solenoid	162
3.4.3.4	Emergency air-shutoff flap - Feedback signal	164
3.4.4	mtu Devices for Emergency Air-Shutoff Flap Control	166
3.4.4.1	Functional description of the equipment	166
3.4.4.2	Extension board AIB 2	167
3.4.4.3	Failure Monitoring Units (FMU1-01 and FMU1-02)	168
3.4.4.4	Failure Monitoring Units (FMU1-01 and FMU1-02) - Technical data	169
3.4.4.5	Microprocessor-controlled high-current relay	170
3.4.5	Start-Up	173
3.4.5.1	SDAF function test	173
3.4.6	Operational Checks	174
3.4.6.1	Weekly test	174

4 Interfaces

4.1	Service Automation Module SAM - Connector assignment	175
4.2	J1939	184
4.2.1	Nodes and baud rate	184
4.2.2	J1939 - Transmitting (mtu to external devices)	185
4.2.3	J1939 - Receiving (external device to mtu)	189
4.2.4	Rolls-Royce Solutions specification for proprietary objects	190
4.2.5	Fault messages J1939 - DM1	192
4.2.6	Special J1939 messages	197
4.3	RK512 interface on RS232	198

5 Cabling

5.1	Plant cable W001 - ECU 7 to SAM	221
5.2	Plant cable W005 - ECU 7 to SAM	224
5.3	Plant cable W007 - Engine to SAM	225
5.4	Plant cable W041 - EMU 7 to SAM	226
5.5	Wiring Harnesses on Engine 4000Px3	228
5.5.1	Engine wiring variants	228
5.5.2	Engine Wiring - Standard Version	229
5.5.2.1	Engine wiring harness - One ECU 7	229
5.5.2.2	Injector wiring harness 12V4000PX3 - One ECU 7	230
5.5.2.3	Injector wiring harness 16V4000PX3 - One ECU 7	231
5.5.2.4	Injector wiring harness 20V4000PX3 - One ECU 7	232
5.5.3	Engine Wiring with redundant ECU 7	233
5.5.3.1	Engine wiring harness - Redundant ECU 7	233
5.5.3.2	Injector wiring harness 12V4000PX3 - Redundant ECU 7	235
5.5.3.3	Injector wiring harness 16V4000PX3 - Redundant ECU 7	236
5.5.3.4	Injector wiring harness 20V4000PX3 - Redundant ECU 7	237
5.5.3.5	Connecting cable MAIN and BACKUP ECU 7 to Switch Box SBX 1	238
5.5.4	Additional Engine Wiring with EMU	239
5.5.4.1	Engine wiring with EMU	239

6 Sensors, Actuators and Injectors on the Engine

6.1	Sensors	240
6.1.1	Sensors and actuators - Overview	240
6.1.2	Pressure Sensors	245
6.1.2.1	Pressure sensor 0 to 6 bar(g)	245
6.1.2.2	Pressure sensor -70 to 70 mbar(g)	249
6.1.2.3	Pressure sensor 0 to 15 bar	253
6.1.2.4	Pressure sensor 0 to 10 bar(g)	257
6.1.2.5	Pressure sensor 0.5 to 4.5 bar(a)	261

6.1.2.6	High-pressure sensor 0 to 2000 bar	265	8.1.1	Engine wiring - Check	320
6.1.3	Temperature Sensors	269	8.1.2	Engine Control Unit - Self-test	321
6.1.3.1	Temperature sensor -40 to 150 °C	269	8.1.3	SAM - Self-test implementation	322
6.1.3.2	Temperature sensor -40 to 900 °C	272	8.1.4	CAN nodes - Configuration	323
6.1.4	Speed Sensors	276	8.1.5	SAM minidialog	325
6.1.4.1	Speed sensor	276	8.1.6	Engine Control Unit - Web feature	329
6.1.4.2	Exhaust turbocharger speed sensor 50 to 100000 rpm	279	8.1.7	Diagnostic lamps of the devices	331
6.1.4.3	Speed sensor	282	8.1.8	Parametrization with dialog unit	333
6.1.5	Level Sensors	285	8.1.9	Fault Code List	335
6.1.5.1	Level sensor	285	8.1.9.1	Fault indication on SAM minidisplay	335
6.1.5.2	Level sensor	289	8.1.9.2	SAM and EMU 7 fault codes	365
6.2	Actuators	293	8.2	Repair Work	374
6.2.1	Injectors	293	8.2.1	SAM - Replacement	374
6.2.2	Limit switch	295	8.2.2	SAM fuse - Replacement	375
			8.2.3	CF card - Replacement	376
			8.2.4	ECU - Replacement	377
			8.2.5	Sensors and actuators - Removal	378
7	Function Area Configurations		9	Appendix A	
7.1	SAM parameters - Overview	296	9.1	List of abbreviations	384
7.2	SAM transistor and PWM outputs - Adjustable	308	9.2	Conversion tables	388
7.3	SAM transistor and relay outputs - Non-adjustable	311	9.3	Contact person/Service partner	392
7.4	SAM analog outputs - Adjustable	312	10	Appendix B	
7.5	SAM analog outputs - Not adjustable	314	10.1	Index	393
7.6	I/O - Configuration	315			
7.7	EMU parameters	317			
8	Repair				
8.1	Diagnosis and Programming	320			

1 Safety

1.1 Important provisions for electronic products

General

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Assembly, maintenance, and repair without ESD („Electro Static Discharge“) equipment and ESD work station
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

1.2 Intended use of electronic products

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:

- Within the permissible operating parameters in accordance with the (→ Technical data)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/Rolls-Royce Solutions contact/Service partner)
- In the original configuration of the delivery or in a configuration approved in writing by the manufacturer
- In compliance with all safety regulations and in adherence with all safety and warning notices in this manual
- Work which requires communication with the control units must only be carried out with a diagnosis software or with tools approved by the manufacturer.
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version is used. The latest version can be found at:

- <http://www.mtu-solutions.com>

Modifications or conversions

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

1.3 Personnel and organizational requirements

Organizational measures of the user/manufacturer

This manual must be issued to all personnel involved in operation, maintenance, repair, assembly, installation, or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair, assembly, installation, and transport personnel at all times.

Personnel must receive instruction on product handling and repair based on this manual. In particular, personnel must have read and understood the safety requirements and warnings before starting work.

This is important in the case of personnel who only occasionally perform work on or around the product. Such personnel must be instructed repeatedly.

Personnel requirements

All work on the product must be carried out by trained, instructed and qualified personnel only:

- Training at the Training Center of the manufacturer
- Qualified personnel from the areas mechanical engineering, plant construction, and electrical engineering and also for work with live parts

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair, assembly, installation, and transport in writing.

Personnel shall not report for duty under the influence of alcohol, drugs or strong medication.

Clothing and personal protective equipment

Always wear appropriate personal protective equipment, e.g. safety shoes, ear protectors, protective gloves, goggles, breathing mask. Follow the instructions concerning personal protective equipment in the respective descriptions of the individual activities or the manufacturer's documentation included in the delivery.

When working on live components, in particular on electrical switchgear or on high-voltage component, wear special protective clothing according to IEC 61482.

Safe handling of Substances of Very High Concern pursuant to the REACH regulation (Registration, Evaluation, Authorization and restriction of CHemicals): We recommend wearing protective gloves at all times in order to reduce risk when working.

1.4 Safety regulations for commissioning and operation of electronic products

Safety regulations for commissioning

Install the product correctly and carry out acceptance in accordance with the manufacturer's specifications before putting the product into service. All necessary approvals must be granted by the relevant authorities and all requirements for initial startup must be fulfilled.

Whenever the product is subsequently taken into operation ensure that:

- All maintenance and repair work has been completed.
- All protective devices are in place.
- The power supply (power plug) must always be accessible to allow isolation of the equipment from supply voltage at any time.
- All components are properly grounded. Ground separately by means of a grounding stake as necessary.
- No persons wearing pacemakers or any other technical body aids are present.
- Adequate ventilation of the operating room must be ensured for any operating status.
- The product must be free of any damage, this applies in particular to lines and cabling.
- Protect battery terminals, generator terminals or cables against accidental contact.
- All connections are correctly arranged/assigned, e.g. polarization.

Immediately after putting the product into operation, make sure that all control and display instruments as well as the monitoring, signaling and alarm systems work properly.

Safety regulations during operation

The operator must be familiar with the control and display elements.

The operator must be familiar with the consequences of any operations performed.

During operation, the display instruments and monitoring units must be observed with regard to present operating status, violation of limit values and warning or alarm messages.

Malfunctions and emergency stop

Practice emergency procedures, especially emergency stopping, at regular intervals.

Take the following steps if any system malfunctions are detected or signaled by the system:

- Inform supervisor(s) in charge.
- Analyze the message.
- If required, carry out emergency operations e.g. isolate the equipment from the power supply.

Contact Service if the root cause of the malfunction cannot be clearly identified.

Operation of electrical equipment

When electrical equipment is in operation, certain components of these appliances are electrically live.

Follow the applicable operating and safety instructions when operating the devices and heed warnings at all times.

1.5 Safety regulations for assembly, maintenance, and repair work on electronic products

Safety regulations for work prior to assembly, maintenance, and repair

Have assembly, maintenance, or repair work carried out by qualified and authorized personnel only.

Never carry out assembly, maintenance, or repair work with the product in operation, unless:

- It is expressly permitted to do so following a written procedure.

Attach "Do not operate" sign in the operating area or to control equipment.

Use the recommended special tools or suitable equivalents when instructed to do so.

Safety regulations when performing assembly, maintenance, and repair work

Special tools and lifting equipment

Use only proper and calibrated tools.

Removing, installing and cleanliness

Pay particular attention to cleanliness at all times.

Working on electrical and electronic assemblies

Always obtain the permission of the person in charge before commencing assembly, maintenance, or repair work and before switching off any part of the electronic system required to do so.

De-energize the relevant areas prior to working on assemblies.

Follow the safety rules:

- Disconnect the electric system from live parts.
- Reliably prevent inadvertent power-on.
- Verify absence of voltage using suitable measuring/test equipment e.g. voltage indicator.
- Connect conductors and grounding with a short-circuiting device, e.g. grounding bar.
- Cover adjacent live parts. Prevent access by unauthorized persons.

ESD (Electrostatic Discharge): Work on components which could be damaged by electrostatic discharge must always be carried out with appropriate equipment. Appropriate equipment are e.g. electrically conductive work surfaces or antistatic wristbands.

Do not damage cabling during removal work. When reconnecting, ensure that cabling cannot be damaged during operation by:

- Contact with sharp edges
- Chafing on components
- Contact with hot surfaces.

Do not secure cables on lines carrying fluids.

Do not use cable ties to secure lines.

Always use connector pliers to tighten union nuts on connectors.

Never use the product as a climbing aid.

Subject the device as well as the product to functional testing on completion of all repair work. The emergency stop function must be tested in particular. The emergency stop function test involves the deactivation of the engine controller power supply. This test may only be performed when the product is cold.

Store spare parts properly prior to replacement, i.e. protect them against moisture in particular. Package faulty electronic components or assemblies properly before dispatching for repair:

- Moisture-proof
- Shock-proof
- Wrapped in antistatic foil (as necessary)